HOCHSCHULE RAVENSBURG-WEINGARTEN UNIVERSITY OF APPLIED SCIENCES

GPIO Output Stages in Simulation

Push-pull, open drain and open collector: switching speed, voltage and current limits

Embedded Computing – Master of Engineering – Worksheet

Overview

Every GPIO pin ends in an output stage: two transistors (push-pull) or one transistor to ground (open drain, open collector). In this worksheet you will take these stages apart in a circuit simulator. You will find out how much current a pin can deliver, how fast it can switch a load, why a pull-up resistor makes a bus slow, and which voltages and currents destroy a pin or the circuit behind it.

Learning outcomes. After this worksheet you can

- model a GPIO output stage by on-resistance and saturation current and relate the model to the V_{OH}/V_{OL} specifications of a data sheet,
- predict rise and fall times, the maximum toggle frequency and the dynamic power of a push-pull output,
- dimension the pull-up resistor of an open-drain bus such as I^2C from rise-time and V_{OL} limits,
- judge where the maximum output voltage and current of a pin come from (body diodes, injection current, back-powering, transistor breakdown and saturation) and protect a driver against inductive kick-back,
- recognise when a PCB trace has to be treated as a transmission line and terminate it.

Organisation. About 4 hours, in teams of two. Tasks marked [P] are preparation on paper and should be done *before* you open the simulation. [S] means simulate and measure, [D] means discuss and explain. Hand in a short report (PDF): for each task your calculation, the measured values (tables, screenshots of the scopes) and your explanation. When a measurement differs from your prediction by more than about $10\,\%$, explain why.

Tool. Falstad CircuitJS runs in the browser: `https://www.falstad.com/circuit/`. Each part of this sheet has a link that opens the prepared circuit. The circuit files are also in the folder `examples/interfaces/falstad/` of the lecture repository (open them with *File* → *Open File*).

Working with CircuitJS

- Hover over an element to see its voltages, currents and power in the lower right corner.
- Click on a logic input (H/L) or a switch to toggle it. Double-click an element to edit its value. Sliders on the right change the values marked in the circuit.
- Scopes: right-click → *Properties* sets the time base ("scope speed"), the scale and shown values (e.g. average, peak-to-peak). Drag with the mouse in a scope to measure time differences.
- *Reset* restarts the simulation, *Run/STOP* freezes it for reading values.
- *Options* → *Other Options* lets you change the simulation time step. If edges look jagged, make the time step smaller.

The transistor model used in all circuits

The MOSFETs follow the square-law (Shichman–Hodges) model with threshold $V_T = 0.6\,\text{V}$ and

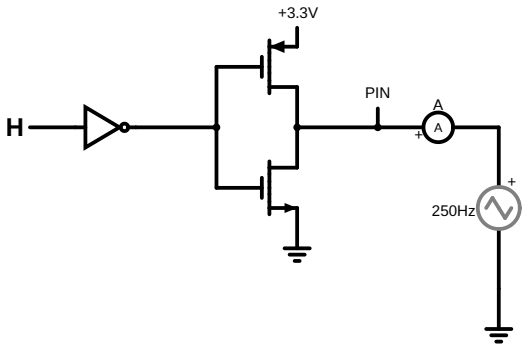
$$I_D = \begin{cases} \beta\left[(V_{GS} - V_T)\,V_{DS} - \tfrac{1}{2}V_{DS}^2\right] & V_{DS} < V_{GS} - V_T \quad \text{(triode)} \\ \tfrac{\beta}{2}\,(V_{GS} - V_T)^2 & V_{DS} \geq V_{GS} - V_T \quad \text{(saturation)} \end{cases}$$

with $\beta_N = 12\,\text{mA/V}^2$ for the NMOS and $\beta_P = 8\,\text{mA/V}^2$ for the PMOS of the output stage (PMOS: use V_{SG}, V_{SD}). Both have a body diode. The supply is $V_{DD} = 3.3\,\text{V}$. The model knows no gate or junction capacitances, no temperature and no breakdown. Pin and load capacitances are drawn as explicit capacitors. Keep in mind what the model cannot show when you interpret the results.

1 Push-pull output: static characteristic

▶ open in Falstad (`A_pushpull_characteristic.txt`)

The logic input is the bit in the output register. Two inverters (the predriver and the CMOS stage) make the pin follow it. A triangle voltage source acts as a curve tracer: it forces the pin voltage from $-0.5\,\text{V}$ to $3.8\,\text{V}$, and an ammeter measures the pin current (positive = current out of the pin). The left scope shows the current over the voltage, the right scope the power in both transistors.



Task 1.1 [P] For $V_{GS} = V_{DD}$, calculate the on-resistance $R_{on} = 1/(\beta(V_{GS} - V_T))$ of both transistors (valid for small V_{DS}) and their saturation currents. Which one is the stronger transistor, and why do chip designers make the PMOS larger than the NMOS?

Task 1.2 [S] Set the output to H and then to L. From the curve, read V_{OH} and V_{OL} at pin currents of 2, 4, 8, 12 and $20\,\text{mA}$ (Run/STOP and hovering over the PIN label helps). Compare with the values from the model equations and enter both in a table. Also read the short-circuit currents (pin at $0\,\text{V}$ for H, pin at $3.3\,\text{V}$ for L).

Task 1.3 [D] The RP2040 data sheet specifies $V_{OH} \geq 2.62\,\text{V}$ and $V_{OL} \leq 0.5\,\text{V}$ at a load current equal to the selected drive strength (2, 4, 8 or $12\,\text{mA}$) and $IOVDD = 3.3\,\text{V}$. Which drive strength setting does the model represent? Drive strength is implemented by switching transistor "fingers" in parallel, i.e. by scaling β. Which β_P would just meet the $4\,\text{mA}$ setting?

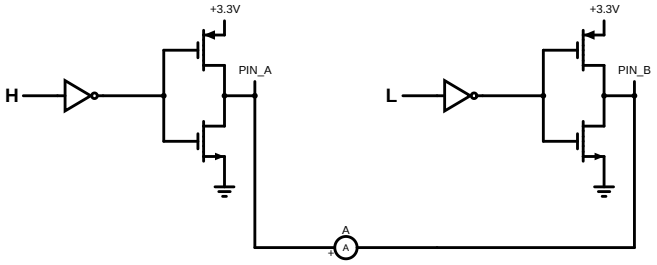
Task 1.4 [P/S] A red LED ($V_F \approx 2.0\,\text{V}$) with a $100\,\Omega$ series resistor is connected from the pin to GND. The naive calculation gives $I = (3.3\,\text{V} - 2.0\,\text{V})/100\,\Omega = 13\,\text{mA}$. Draw the load line $I = (V_{PIN} - V_F)/R$ into the measured characteristic and find the real operating point. Optional: verify it by replacing the sweep source with an LED and a resistor (*Draw → Outputs and Labels → Add LED*).

Task 1.5 [D] Look at the curve below 0 V and above 3.3 V. Which element conducts there? Relate this to the absolute maximum rating "pin voltage $-0.5 \ldots IOVDD + 0.5$ V" of the RP2040 and to the term *injection current*. Why can a large injection current upset an analogue measurement or even a digital circuit on the same chip?

Task 1.6 [S/D] Read the maximum power in the transistors from the right scope. Many microcontrollers allow 50 mA to 100 mA in total for all pins of one supply domain. What does this mean for the number of pins that may drive LEDs at the same time, and what happens to the chip temperature if a pin is shorted to GND?

2 Two push-pull outputs on one line

▶ **open in Falstad** (`C_bus_contention.txt`)



Task 2.1 [P] Output A drives high, output B drives low. Set up the equation $I_{D,P,A}(V) = I_{D,N,B}(V)$ for the line voltage V and solve it numerically. Which region does each transistor work in? Calculate the current and the power in both transistors.

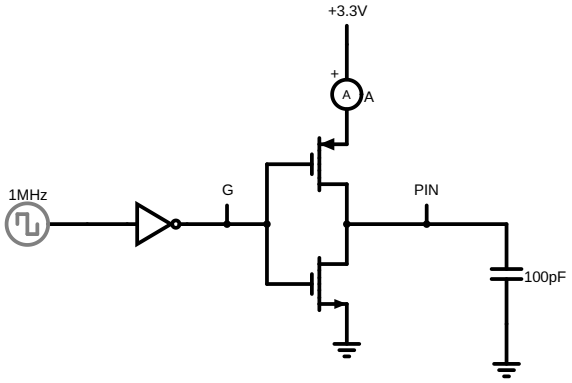
Task 2.2 [S] Verify with the simulation. What logic level does a receiver see on the line?

Task 2.3 [D] Name three situations in embedded practice where this happens. How do tri-state outputs, open-drain outputs and series resistors prevent the damage? Calculate the series resistor in each output that limits the contention current to 5 mA.

3 Push-pull output: switching speed and dynamic power

▶ open in Falstad (`B_pushpull_capacitive_load.txt`)

The square wave is the program toggling the pin. The predriver is a logic inverter with an adjustable slew rate (how fast the gate voltage changes). The output drives the load capacitance C_L (pin, trace, inputs of other chips). An ammeter measures the current taken from the supply.



Task 3.1 [P] Estimate the rise and fall times (10 % to 90 %) for $C_L = 10$ pF, 100 pF and 1 nF:

(a) with the RC approximation $t_r = 2.2\, R_{on} C_L$,

(b) exactly with the square-law model: the transistor starts in saturation (constant current) and enters the triode region when $V_{DS} = V_{GS} - V_T$. Integrate $dt = C_L\, dV / I_D(V)$. Hint: $\int \frac{dx}{x(2a-x)} = \frac{1}{2a} \ln \frac{x}{2a-x}$.

Task 3.2 [S] Set the predriver slew rate to its maximum and measure t_r and t_f for the three capacitances (set the scope time base accordingly, or zoom in with Run/STOP). Compare with both predictions. For which C_L does the predriver dominate the edge?

Task 3.3 [S/D] Increase the toggle frequency with $C_L = 1$ nF until the pin no longer reaches valid logic levels ($V_{IH} = 0.7\, V_{DD}$, $V_{IL} = 0.3\, V_{DD}$). Derive a rule of thumb for the maximum frequency from t_r and t_f. Why do data sheets specify the maximum GPIO frequency only together with a load capacitance?

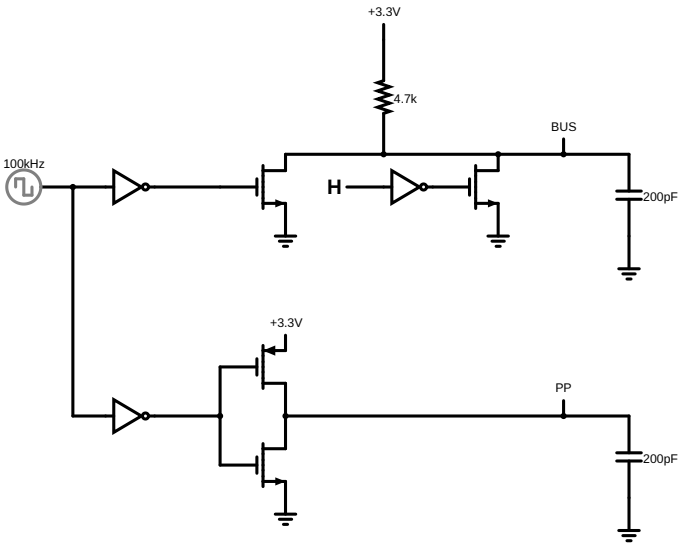
Task 3.4 [S] Set $f = 1$ MHz and the fastest slew rate. Read the average supply current (right scope) for the three capacitances and compare with $I = C_L V_{DD} f$. Show that the energy taken from the supply per cycle is $C_L V_{DD}^2$, independent of R_{on}. Where is it dissipated?

Task 3.5 [S/D] Set $C_L = 1$ pF and reduce the predriver slew rate to 0.02 V/ns. Observe the supply current. Explain what you see, calculate the time window in which both transistors conduct, and compare the average supply current with $C_L V_{DD} f$. Why do microcontroller input pins have Schmitt triggers, and why does a slowly changing analogue voltage on a digital input increase the current consumption?

4 Open-drain bus versus push-pull

▶ open in Falstad (`D_open_drain_bus.txt`)

Two open-drain devices share a line with one pull-up resistor R_P, as on an I²C bus. Device 1 sends a clock; device 2 can hold the line low. A push-pull stage drives the same capacitance for comparison.



Task 4.1 [P] Show that the I²C rise time (30 % to 70 % of V_{DD}) of an RC charge is $t_r = \ln(7/3)\, R_P C_b = 0.8473\, R_P C_b$. Calculate it for $R_P = 4.7\,\text{k}\Omega$ and $C_b = 200\,\text{pF}$.

Task 4.2 [S] Measure the rise and fall times of BUS and the rise time of PP (same thresholds). Explain the difference between rising and falling edge of the open-drain line.

Task 4.3 [P] Dimension R_P for $C_b = 200\,\text{pF}$ and $V_{DD} = 3.3\,\text{V}$ according to the I²C specification (NXP UM10204): $V_{OL} \le 0.4\,\text{V}$ at a sink current of 3 mA, $t_r \le 1000\,\text{ns}$ (Standard-mode, 100 kHz) and $t_r \le 300\,\text{ns}$ (Fast-mode, 400 kHz). Give the permitted range of R_P for both modes.

Task 4.4 [S] Check your limits in the simulation: measure t_r at $R_{P,max}$ for Fast-mode, and V_{OL} and the current through R_P during the low phase for $R_P = 4.7\,\text{k}\Omega$, $1\,\text{k}\Omega$, $470\,\Omega$ and $220\,\Omega$. Increase C_b to 400 pF (the maximum allowed by I²C) with $R_P = 4.7\,\text{k}\Omega$: which mode still works?

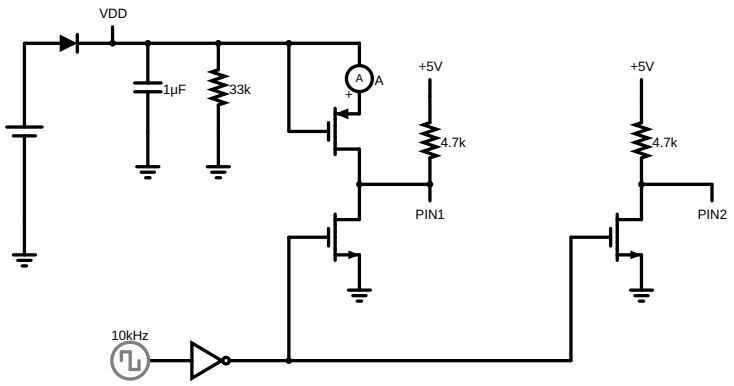
Task 4.5 [S/D] Set the input of device 2 to L. What happens to the clock of device 1? Which I²C feature uses this behaviour? What would happen with two push-pull outputs (part 2)?

Task 4.6 [D] Calculate the static power in R_P while the line is low, and the product of this power and the rise time. What follows for the trade-off between speed and power of an open-drain bus, compared to push-pull? Propose a circuit that speeds up the rising edge without reducing R_P, and try it in the simulation (e.g. a current source pull-up or a switched pull-up).

5 Maximum output voltage: open drain with a 5 V pull-up

▶ open in Falstad (`E_open_drain_overvoltage.txt`)

A 3.3 V microcontroller drives a 5 V logic input through an open-drain output and a pull-up to 5 V. PIN1 is a normal GPIO in open-drain mode: the PMOS is switched off, but it is still there. PIN2 is a 5 V-tolerant open-drain pin without a PMOS. The 3.3 V regulator can only source current (diode). The resistor "MCU load" represents the current consumption of the chip.



Task 5.1 [P] Predict the high level of PIN1 and PIN2 and the current that flows from the 5 V supply into the 3.3 V supply via PIN1 ($R_{PU} = 4.7\,\text{k}\Omega$, diode forward voltage about 0.7 V).

Task 5.2 [S] Measure both high levels and the injection current. Is the high level of PIN1 sufficient for a 5 V CMOS input with $V_{IH} = 0.7 \cdot 5$ V?

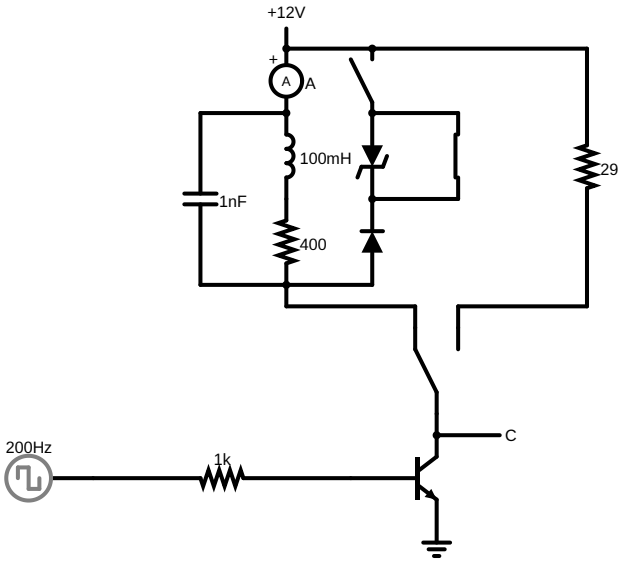
Task 5.3 [S/D] Set $R_{PU} = 1\,\text{k}\Omega$ and keep the MCU load at $33\,\text{k}\Omega$ ($100\,\mu$A, e.g. in a sleep mode). Observe V_{DD}. Explain the effect (*back-powering*). What happens if the microcontroller is switched off ($V_{DD} = 0$) but the 5 V side stays on?

Task 5.4 [D] Look up in the data sheets of an RP2040, an RP2350 and an STM32F4 which pins are 5 V tolerant and how large the injection current per pin and in total may be. Name two circuits that shift a 3.3 V open-drain signal to 5 V correctly, one of them bidirectional (for I^2C).

6 Open collector: high voltage, high current, inductive loads

▶ open in Falstad (`F_open_collector_inductive.txt`)

A 3.3 V GPIO switches an NPN transistor ($\beta = 100$) through the base resistor R_B. The collector is an open-collector output for a 12 V load: a relay coil (0.1 H, $400\,\Omega$, 1 nF winding capacitance) or a 12 V/5 W lamp. The switch *flyback* adds a free-wheeling diode, opening the switch *bypass* puts a 24 V Zener diode in series with it. The timescale is shortened (switching at 200 Hz) so that you can see several cycles.



Task 6.1 [P] Calculate the coil current, the base current with $R_B = 1\,\text{k}\Omega$ and the overdrive factor $\beta I_B / I_C$. Is the transistor saturated?

Task 6.2 [P/S] Without flyback diode: when the transistor switches off, the coil current continues and charges the winding capacitance. Estimate the peak collector voltage from energy conservation ($\frac{1}{2}LI^2 = \frac{1}{2}CV^2$) and measure it. Compare with V_{CEO} of a BC547 (45 V) or of a ULN2003 driver (50 V). What does the simulation not show?

Task 6.3 [S] Close *flyback*. Measure the maximum collector voltage and the time until the coil current has fallen to $10\,\%$ (right-click the scope, *Properties*, to change the time base). Then open *bypass* (diode + 24 V Zener) and measure again. Calculate both times ($\tau = L/R$) and explain why the Zener variant is faster. Why does that matter for a relay?

Task 6.4 [S/D] Increase R_B to $47\,\text{k}\Omega$. Measure I_C and V_{CE}. Will the relay still pull in?

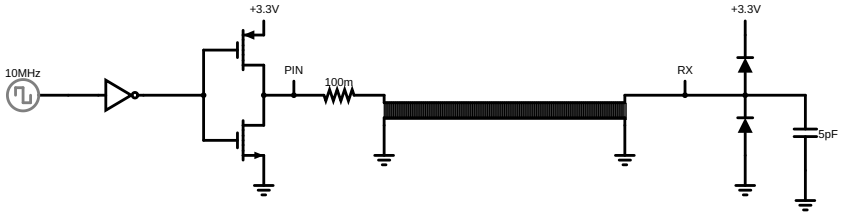
Task 6.5 [S/D] Switch the load to the lamp. Measure I_C, V_{CE} and the power in the transistor for $R_B = 1\,\text{k}\Omega$ and $330\,\Omega$. A small-signal transistor in SOT-23 tolerates about 250 mW. Why is $330\,\Omega$ still not a good solution? Propose a better driver.

Task 6.6 [D] Summarise: what limits the maximum output voltage and the maximum output current of a push-pull pin, of an open-drain pin and of an external open-collector stage?

7 Fast edges on a PCB trace

▶ open in Falstad (`G_transmission_line.txt`)

The push-pull pin drives a $50\,\Omega$ trace with 1 ns delay (about 15 cm on FR4). At the end there is the input of another chip: 5 pF and ESD diodes to V_{DD} and GND. R_S is a series resistor at the driver. The output stage is set to its highest drive strength and fastest slew rate, modelled by $\beta_N = 30$ mA/V^2 and $\beta_P = 25$ mA/V^2. The slew rate is adjustable as on many microcontrollers (RP2040: `SLEWFAST`, STM32: `OSPEEDR`).



Task 7.1 [P] A trace is "electrically long" if the edge is shorter than about twice the delay of the trace. Is that the case here with the fastest slew rate? Calculate the reflection coefficients at the receiver (open end) and at the driver ($R_{on} + R_S$ against Z_0), the voltage step launched into the line and the first voltage at the receiver, for $R_S = 0$.

Task 7.2 [S] Measure overshoot and undershoot at RX and the peak current in the ESD diodes for $R_S = 0$. Then find the R_S that gives the cleanest edge at RX. Compare with the rule $R_S = Z_0 - R_{on}$.

Task 7.3 [S/D] Set R_S back to 0 and reduce the slew rate. From which slew rate on does the ringing disappear? Why should you select the slowest slew rate that meets the timing of your interface?

8 Design task

Task 8.1 [P/S/D] A 3.3 V microcontroller board has to

(a) read a 5 V sensor over I^{2}C at 400 kHz; the cable is 30 cm long, the total bus capacitance is estimated at 250 pF,

(b) switch a 24 V/50 mA solenoid valve at up to 10 Hz,

(c) send a 10 MHz SPI clock over a 10 cm trace to a flash memory.

For each signal choose the output type and configuration of the pin (push-pull/open drain, drive strength, slew rate), dimension the external components and verify your choice in a simulation that you build from the given circuits. Document the remaining risks.